INR-Based Anticoalugant Dose Optimization System

PRESENTED BY

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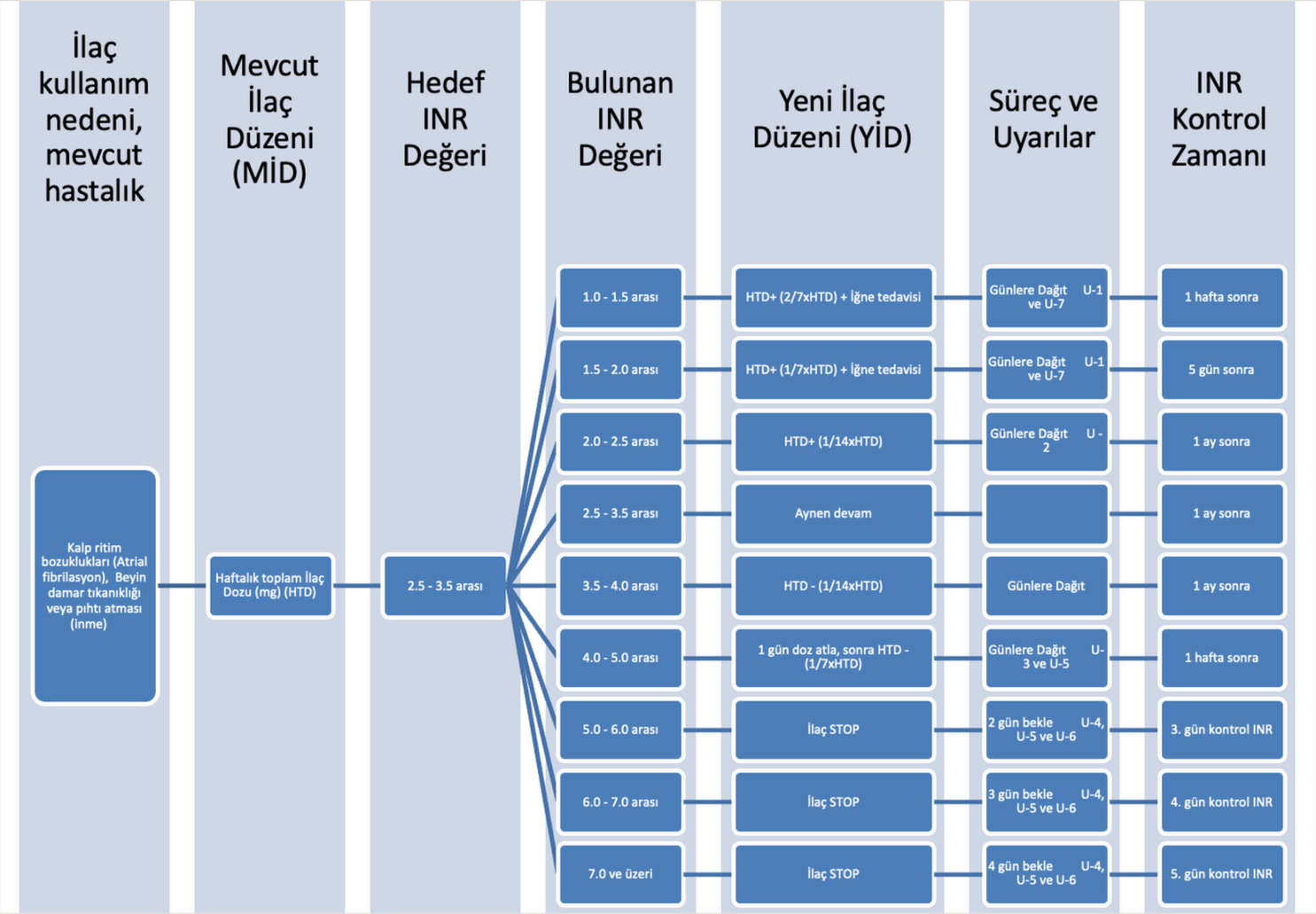
Project Overview

- INR measures blood clotting level
- Warfarin is an anticoagulant that requires precise dose management
- Patients must keep INR within a target range to avoid risks
- This system supports dose adjustment and decision-making

Planned System Design

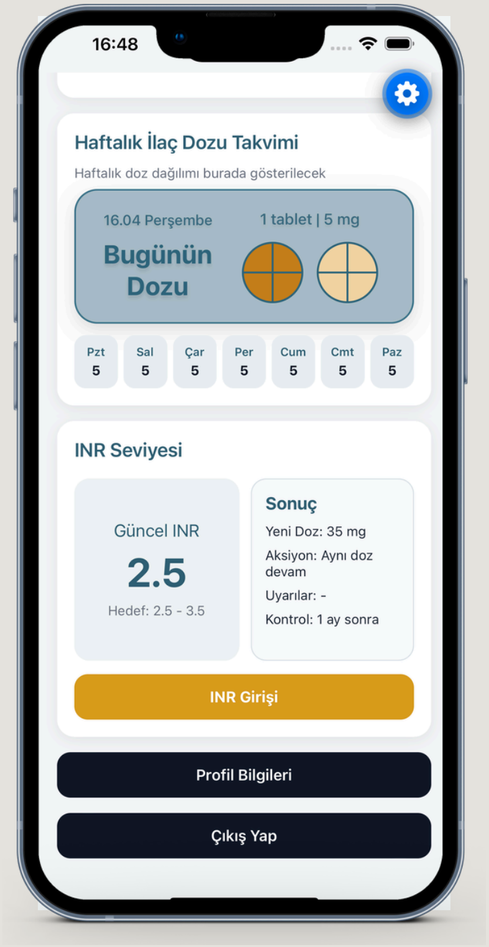
- Designed for simple and intuitive use
- Focus on elderly-friendly interface
- Visual weekly dose tracking

Clinical Decision Logic



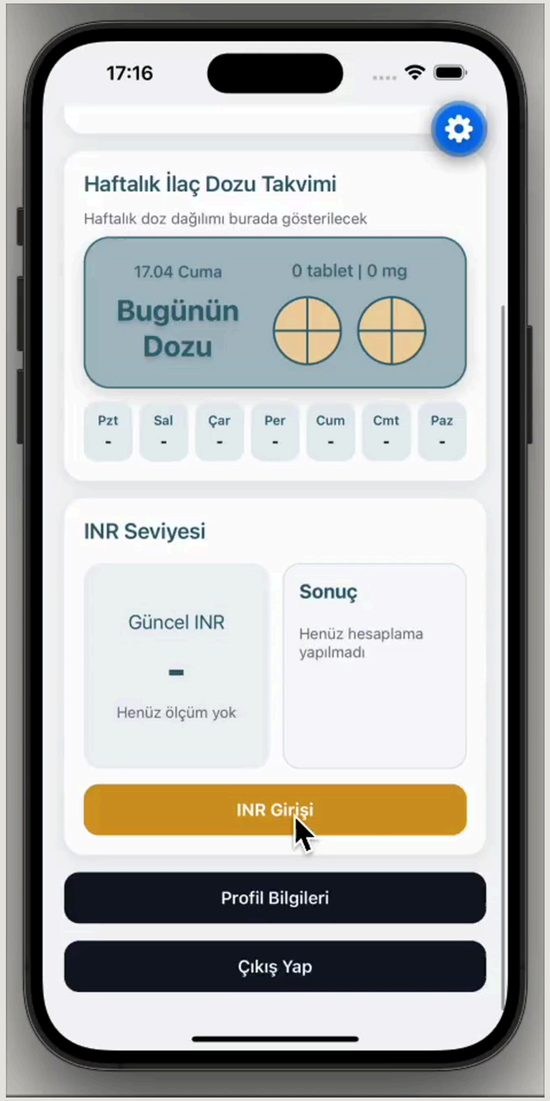
- Based on clinical INR guidelines
- Adjusts weekly dose (increase / decrease / stop)
- Provides warnings and next control timing

What I Built So Far



- Set up project structure and navigation
- Implemented user authentication (Firebase)
- Developed INR input system (Picker-based)
- Built initial rule-based dosing algorithm
- Implemented weekly dose calculation logic
- Created basic UI screens and data flow

How the System Works



INR Input

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Dose Calculation

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Weekly Dose

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Daily Plan

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UI Output

Current Status

- Core system structure is established
- Rule-based dosing logic implemented (AF scenario)
- INR input and calculation flow working
- Basic UI connected to algorithm

Next Steps

- Extend algorithm to other clinical cases
- Add interaction factors (medication & diet)
- Move logic to backend (Cloud Functions)
- Improve UI and usability

MUĞLA SITKI KOÇMAN ÜNİVERSİTESİ

Thank You

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